SPARKLAB
(HarvestMind)

PRODUCT REQUIREMENT DOCUMENT (PRD)

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umhackathon@um.edu.my

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Project Name: HarvestMind

Version: 1.0

1. Project Overview

- **Problem Statement:** Malaysians farmers rely on traditional planting cycles without accounting for market saturation or weather volatility, leading to unexpected losses when crop prices crash at harvest time. They lack access to integrated tools that combine economic, environmental, and social signals to support smarter decisions.
- **Target Domain:** Agricultural decisions intelligence for smallholder farmers in Malaysia, specifically targeting crop planning, market timing, and risk mitigation in the agricultural sector.
- **Proposed Solution Summary:** HarvestMind is a GLM-powered Decision Intelligence Engine that ingests structured data (historical prices, weather patterns) and unstructured signals (regional news, pest outbreaks, trade policies) to perform Pivot vs Perservere analysis, recommending whether farmers should continue, switch crops, or harvest early with specific, actionable guidance.

2. Background & Business Objective

- **Background of the Problem:** For generations, Malaysian farmers have depended on intuition and seasonal tradition to make planting and harvesting decisions. Without access to real-time market intelligence or weather-risk tools, they invest months of labor and significant capital only to face price crashes driven by oversupply. When too many farmers plant the same crop simultaneously, market saturation drives prices below the cost of production. This systemic issue affects not just individual income, but also local food supply stability and national agricultural productivity.

- **Importance of Solving This Issue:** Economic losses from poor crop timing affect rural communities disproportionately. A solution that provides timely, data-grounded

recommendations can prevent income loss, reduce food waste, stabilize local markets, and improve food security. Democratizing access to decision intelligence tools, previously available only to large agribusinesses, empowers smallholders farmers to compete and thrive.

- **Strategic Fit / Impact:** HarvestMind aligns with Malaysia's National Agrofood Policy and the national agenda for agricultural modernisation. By leveraging ILMU GLM, Malaysia's own AI model, alongside a robust data pipeline (FAMA, Open-Meteo, GDELT, Firecrawl), the system supports local AI adoption, strengthens rural digital inclusion, and contributes to national food security goals.

3. Product Purpose

3.1. Main Goal of the System: To provide Malaysian farmers with a full-stack AI-powered decision intelligence platform that delivers clear, actionable crop management recommendations grounded in live market prices, real-time weather data, and regional news signals, enabling confident, data-driven farming decisions.

3.2. Intended Users (Target Audience):

- Smallholder crop farmers in Malaysia
- Agricultural cooperatives and farmers groups
- Local agricultural extension officers
- Policy makers and agri-subsidy planners
- Rural agri-entrepreneurs and crop traders

4. System Functionalities

4.1. Description:

HarvestMind operates as a modular full-stack platform. The frontend (Vanilla JS/HTML) communicates with a Node.js proxy server, which routes requests to a FastAPI orchestrator backend. The backend integrates ILMU GLM for reasoning, Open-Meteo for weather data, FAMA/Firecrawl for live market prices, and GDELT for news intelligence. PostgreSQL stores farmer profiles and decision history, while Celery and Redis handle background market signal generation. The entire stack is containerised via Docker Compose.

4.2. Key Functionalities:

- **Conversational Farm Advisor:** Farmers interact via natural language chat to ask questions about crop timing, market conditions, pest risks, and planting strategy. Powered by ILMU GLM 5.1 via FastAPI.

- **Structured Analysis Engine:** A side panel allows structured input of crop type, GPS location, current prices, weather data, humidity, rainfall, and regional news for a comprehensive AI-driven analysis.

- **Weather Risk Integration:** Open-Meteo API provides real-time rainfall forecasts, temperature, and humidity data based on the farmer's GPS location for weather-aware recommendations.

- **News & Policy Signal Processing:** GDELT news intelligence and Firecrawl web search surface regional pest outbreak reports, government policy changes, and international tariff news as decision inputs.

- **Pivot vs. Persevere Decision Engine:** Every analysis returns a clear recommendation with specific percentage-level actions such as 'Shift 30% labor to ginger to hedge against chili price drop.

- **Persistent Farmer Profiles:** PostgreSQL database stores farmer location, crop history, and past decisions for continuity across sessions.

- **GPS-Based Location Intelligence:** Browser GPS detects location using OpenStreetMap reverse geocoding, feeding regional context into both weather and market recommendations.

- **Market Reference Dashboard:** A live-updating markets page shows Malaysian crop price ranges, 30-day trends, and Buy/Hold/Watch/Sell signals sourced from live FAMA data.

- **Crop Reference Guide:** A filterable guide covering 12 major Malaysian crops with seasonal windows, risk levels, and market characteristics.

4.3. AI Model & Prompt Design

4.3.1. Model Selection: HarvestMind uses ILMU GLM 5.1, Malaysia's own large language model built on the GLM architecture. GLM was selected for: (1) strong multilingual reasoning supporting both English and Bahasa Malaysia; (2) structured output reliability suited for decision-support recommendation formats; (3) compatibility with the ILMU.ai API which integrates directly into the FastAPI orchestrator; and (4) alignment with the hackathon's requirement to use Z AI GLM.

4.3.2. **Prompting Strategy:** HarvestMind uses a multi-step agentic prompting strategy. The FastAPI orchestrator assembles a rich context prompt combining the farmer's location, crop type, live FAMA prices, Open-Meteo weather data, and GDELT news signals before passing to GLM. This few-shot context priming ensures grounded, specific recommendations. The system always concludes with a Pivot/Persevere/Harvest Early decision and percentage-level action guidance.

4.3.3. Context & Input Handling: The system accepts up to 1,000 characters per free-text message and structured inputs across 10 form fields. The FastAPI backend compiles all data

sources, live prices, weather, news, farm profile, into a single enriched prompt with a max token limits of 1000 output tokens. Inputs exceeding character limits are truncated at the input layer. Oversized API responses from data sources are chunked by the backend before being injected into the GLM prompt context.

4.3.4. Fallback & Failure Behavior: When GLM returns an unusable response, the frontend displays a graceful error message. The FastAPI backend implements a retry mechanism with exponential backoff for transient API failures. If live FAMA or weather data is unavailable, the system falls back to cached data from PostgreSQL with a staleness warning displayed to the user. The Firecrawl web intelligence fallback activates automatically when primary data APIs fail.

5. User Stories & Use Cases

User Stories:

- As a chili farmer in Kelantan, I want to know if I should harvest early this month so that I can avoid the price crash when oversupply hits the market.

- As a smallholder farmer, I want to enter my current crop, price trend, and weather conditions so that HarvestMind gives me a clear Pivot or Persevere recommendation based on live data.

- As a farmer cooperative officer, I want to see live market price signals for ginger and durian from FAMA so that I can advise members on the best time to sell.

- As a rural farmer unfamiliar with technology, I want to simply chat in plain language about my farm situation so that I receive understandable, actionable advice.

- As a policy maker, I want to see automated weekly market signal cards so that I can direct subsidies to the most vulnerable farming communities based on real data

Use Cases (Main Interactions):

Free Chat Flow: Farmer opens HarvestMind, types a question in natural language. The Node.js proxy forwards the request to FastAPI, which enriches it with live FAMA prices and Open-Meteo weather data before passing to GLM. The response includes a clear Pivot / Persevere / Harvest Early recommendation.

Structured Analysis Flow: Farmer clicks 'Run Analysis', fills in crop type, GPS location, current price, and news signals. FastAPI compiles this with live weather and market data into an enriched GLM prompt and returns a detailed, farm-specific recommendation.

Market Dashboard Flow: Farmer navigates to the Markets page to check live FAMA prices, 30-day trends, and automated signal cards generated by Celery background workers.

Background Intelligence Flow: Celery workers run weekly to scrape FAMA prices, pull GDELT news signals, and generate market risk/opportunity cards stored in PostgreSQL and displayed on the Markets page.

6. Features Included (Scope Definition)

• Conversational AI chat interface powered by ILMU GLM 5.1 via FastAPI orchestrator

• Live real-time crop price scraping from FAMA via self-hosted Firecrawl web intelligence

• Automated weekly market signal card generation via Celery background workers

• PostgreSQL database for persistent farmer profiles, crop history, and decision records

• Redis task queue for background market data processing and caching

• Open-Meteo weather API integration for real-time rainfall, temperature, and forecast data

• GDELT news intelligence for regional pest outbreak and trade policy signal processing

• GPS-based location detection with OpenStreetMap reverse geocoding

• Structured farm analysis panel with 10 input fields (crop, price, weather, news)

• Pivot vs. Persevere decision engine with percentage-level action guidance

• Market Intelligence dashboard with live Malaysian crop prices and Buy/Hold/Watch/Sell signals

• Malaysian Crop Reference Guide covering 12 crops with filterable categories

• Dark mode toggle with localStorage preference persistence

• Mobile-responsive design with hamburger navigation menu

• Secure Node.js proxy server for API key management

• Full Docker Compose containerised deployment (FastAPI + PostgreSQL + Redis + Firecrawl + Node.js)

• Firecrawl web intelligence fallback for when primary data APIs are unavailable

7. Features Not Included (Scope Control)

• User authentication, login system, or multi-user account management

• Multi-language support , system currently operates in English only

• Offline mode or Progressive Web App (PWA) capability

• Push notifications for real-time market alerts or price threshold warnings

• Integration with official government agricultural databases beyond FAMA

• Mobile native app (iOS/Android) , web-based only

• Direct payment or marketplace features for crop trading

• Multi-user simultaneous sessions with shared farm data collaboration

• Automated crop disease image recognition or computer vision features

8. Assumptions & Constraints

LLM Cost Constraint: Each average user session consumes approximately 1,500–3,000 tokens including the enriched context from live FAMA prices, Open-Meteo weather data, and GDELT news signals. To keep inference costs manageable, the FastAPI backend caches repeated market data queries in Redis with a 1-hour TTL, reducing redundant data API calls. The system limits max_tokens to 1,000 per GLM response and maintains only the last 10 conversation turns in the prompt context.

Technical Constraints: The full stack requires Docker and Docker Compose to run locally. Cloud deployment requires separate services for PostgreSQL, Redis, FastAPI, and Node.js. The Firecrawl self-hosted instance requires its own container and may consume significant memory. FAMA website scraping may break if the site structure changes.

Performance Constraints: Response latency includes FAMA scraping time (2–5 seconds), Open-Meteo API call (0.5–1 second), GDELT query (1–3 seconds), and GLM inference (3–8 seconds), totaling up to 15 seconds for a full structured analysis. Free-tier cloud deployment may experience cold-start delays. Background Celery workers reduce real-time scraping load by pre-computing market signals.

User Input: The AI model requires human-initiated prompts and does not autonomously generate recommendations. All recommendations are AI-generated and should be cross-referenced with local agricultural officers. The system includes a disclaimer on every page.

Geographic Constraint: HarvestMind is designed specifically for Malaysian agricultural conditions. Crop data, FAMA market prices, weather assumptions, and policy context are Malaysia-specific. Use in other geographic contexts may produce inaccurate recommendations. **Data Dependency:** The system's accuracy depends on the availability and reliability of FAMA, Open-Meteo, and GDELT services. Downtime or structural changes to these external sources may affect data quality. Redis caching and PostgreSQL fallback mitigate short-term outages

9. Risks & Questions Throughout Development

• **Recommendation Accuracy:** How do we ensure HarvestMind's Pivot / Persevere recommendations are grounded in actual Malaysian market realities? The system mitigates this by injecting live FAMA prices, Open-Meteo weather data, and GDELT news signals directly into the GLM prompt context. Future versions should validate GLM outputs against known crop price ranges to flag anomalous recommendations.

• **FAMA Scraping Reliability:** What happens if FAMA changes its website structure and Firecrawl scraping breaks? The system uses Redis-cached data as a fallback and displays a staleness warning to users. A more robust solution would integrate with an official FAMA API if one becomes available.

• **API Key Security:** How do we prevent exposure of the ILMU GLM API key? All API calls are routed through the Node.js proxy and FastAPI backend, the key is never exposed to the browser. Environment variables via Docker and Railway/Render environment configs keep keys out of the codebase.

• **GPS Data Privacy:** What happens to farmer location data? Coordinates are used only for the duration of the session for weather lookups and are not stored beyond the PostgreSQL farmer profile with user consent. No persistent tracking occurs without explicit permission.

• **Model Hallucination Risk:** What if GLM generates plausible-sounding but factually incorrect market advice? The FastAPI backend validates GLM outputs against known crop price ranges from the PostgreSQL database. Out-of-range recommendations trigger a fallback response. All pages include a disclaimer to verify advice with local agricultural officers.

• **Infrastructure Complexity:** The full Docker Compose stack (FastAPI + PostgreSQL + Redis + Firecrawl + Node.js) is complex to deploy and maintain. This is mitigated by the included doctor.sh diagnostic script and comprehensive AGENTS.md development guide. Cloud deployment on Railway simplifies orchestration.

• **Mobile Usability in Rural Areas:** Will farmers with older smartphones in low-connectivity rural areas be able to use HarvestMind? The frontend is mobile-responsive and lightweight, but the full backend analysis may be slow on 3G connections. A future offline-capable PWA version with pre-cached crop data would improve rural accessibility